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Teacher and Teaching-related Practices Student Teachers May Have Difficulty Believing They Can Enact/Engage in While in Field Practice: Perspectives from Student Teachers and Teachers in Field-practice Schools

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ABSTRACT
Field practice is pivotal in teacher education, serving both as a bridge connecting the theoretical and practical dimensions of the teaching profession in real-world schools, and as a place for acquiring the complex core skills of the profession in their own right. While prior work has mainly explored general teacher self-efficacy, or broad student-related concerns in field practice, there is surprisingly little knowledge of the specific teacher and teaching-related practices that might challenge student teachers during field practice. This study aims to address this gap by exploring the issue both from the student teacher and the field-practice teacher perspective. Utilizing focus groups and individual interviews with students and field-practice teachers, the study employs content analysis, and an innovative agreement circle to identify which practices students may doubt they can enact/engage in while in field practice. The findings highlight four main areas of concern stemming from both students and field practice teachers: adaptability in teaching; balancing subject knowledge with pedagogical competence; classroom leadership; and teaching independently. The study also uncovers discrepancies between student teachers, who are preoccupied with their teaching performance, and field-practice teachers, who also emphasize the importance of collaborative practices with, for example, parents. The study opens a new field of inquiry and is the first step towards a theory of practice self-efficacy in education. Future research could focus on creating measurement tools to evaluate student teacher efficacy in field practice within the four identified areas, thereby enabling identification of early challenges, assessment of field-practice effectiveness, and the provision of tailored field-practice preparation.

Keywords: *practice self-efficacy, teacher education, field practice in schools, student teachers, field-practice teachers*

1. Introduction

Field practice is an integral and necessary part of teacher education, as it provides student teachers with opportunities to learn the profession's core skills and practices within a real-life professional work environment (Caries & Almeida, 2005). The core skills and profession-specific practices undertaken in field practice, are practiced and learned through increasing involvement and enactment during the program and field practice and will most likely also increase in complexity as the program progresses. What is actually at stake is student teachers learning professional skills while practicing them, and thus, inevitably, student teachers' identity will be split between being a student teacher and being a professional (e.g., Anspal et al., 2018; Alstrup, 2006). This duality is not in play during on-campus teaching. In this sense, the field practice learning environment is relatively unsafe, psychologically, and high risk – in comparison to the on-campus learning environment – since the consequences of "failure" can extend to other people, i.e., the children and their parents, as well as "teacher colleagues". The personal psychological risk is also higher in the profession's learning environment, compared to the on-campus environment, since student teachers face increased communicative, emotional, and psychological demands while learning (Anspal et al., 2018; Caries & Almeida, 2005). Thus, the nature of the learning environment in field practice may lead to student teachers doubting whether they have what it takes to engage in the profession's core practices (Caries & Almeida, 2005).

1.1. Students' doubts about their abilities in relation to field practice

Such doubts have previously been investigated as a matter of self-efficacy. The concept of self-efficacy, which was originally developed by Albert Bandura (1977), covers the belief in one's own capability to plan and perform necessary actions to attain a certain outcome. This stands in contrast to outcome expectations, which cover the expected effects of the performance, such as social recognition, financial gain, salary increase, health effects, professional recognition, job promotion, etc. Self-efficacy is thus a question of how well an individual expects they will be able to perform certain actions or tasks in a specific situation or context, given their capabilities. Self-efficacy is not determined by the skills you have, "but with what you believe you can do with what you have under a variety of circumstances" (Bandura, 1997, p. 37). Variation in efficacy beliefs can have a large impact on how persons perform in a given context, even though they have the same set of skills. (Bandura, 1997). This means that highly skilled individuals will likely be less successful, if they have low self-efficacy with regard to the contextual tasks, compared to persons with higher self-efficacy.

While teachers' self-efficacy is one of the few individual characteristics that reliably predicts instructional practice and student outcomes (Brown et al., 2021), we have not been able to locate any studies specifically targeting or measuring student teacher self-efficacy in relation to field-practice teaching and engaging in specific teacher and teaching-related practices. Previous research has looked into teacher self-efficacy in relation to field practice placements, using instruments intended for the measurement of self-efficacy for teaching with trained teachers, not student teachers. For example, Woolfolk Hoy and Spero (2005) used a 10-item version of the Teacher Efficacy Scale (Gibson & Dembo,1984), and Bandura's (1997) 30-item Teacher Self-Efficacy Scale, to identify significant predictors of preservice teachers' sense of efficacy: (a) the relationship between preservice teacher and mentor supervisor; (b) the quality of support from the cooperating teacher, the school community, and the field placement supervisor; and (c) the number of field experiences. Flores (2015) examined the general science teaching self-efficacy and personal science teaching efficacy of student teachers before and after a five-week placement at a public elementary school, using the Science Teaching Efficacy Belief Instrument-Preservice (Enochs & Riggs, 1990), based on Bandura's theory of self-efficacy (1977). Flores found that both the general and the personal science teaching self-efficacy increased significantly and substantially from the pre to the post measurement. Li and Zhang's (2000) results were diverse, as they found that early field practice experiences affected student teachers' general teaching self-efficacy negatively, but their personal teaching self-efficacy positively, from before to after field experience, as measured with the Teacher Efficacy Scale by Hoy and Woolfolk (1990). Mok et al. (2023) conducted a metanalysis of 28 studies that evaluated the effect of intervention studies for preservice teachers and beginning teachers on their teacher self-efficacy. They found an overall medium effect size (Cohen's d) for both groups, and the effect was amplified when interventions included feedback on lesson plans.

With a broader scope than teacher self-efficacy, Anspal et al. (2018), through interviews with Estonian student teachers, found that a source of tension was the relationship between self-conception and the professional role they were entering as student teachers, also expressed as the difference between teaching practice and the university part of their education. This tension and doubt varied over time for different students. Thus, for some the tension grew as teaching demands increased with more challenging pupil groups, such as special-needs pupils. Other student teachers expressed little or no tension, particularly those who had their teaching practice in schools with a particularly supportive school culture. Likewise, in a study of Malaysian student teachers' concerns during field practice, Goh and Matthews (2011) used reflective journals written by 14 student teachers to document their teaching experiences, concerns, and confidence to teach during field practice, deriving 17 concerns related to their field practice. The field practice related concerns were grouped into four themes: classroom management and student discipline (e.g., concerns about a lack of discipline among students); institutional and personal adjustments (e.g., concerns about meeting expectations of the field practice teacher); classroom teaching (e.g., concerns about organization of teaching activities); and pupil learning (e.g., concerns about capturing student's interest and attention). While both Anspal et al. (2018) and Goh and Mathews (2011) examine concerns and tensions related to student teachers' field practice, more specific concerns were derived from journals recorded while in field practice. These concerns were derived over time and during several field practices. Thus, neither study examined in detail which core practices the student teachers prior to field practice thought could be teacher or teaching-related practices they would doubt they had what it takes to enact or engage in while in field practice, that is, self-efficacy beliefs in this regard.

In Nordic research, there has been a considerable number of studies examining field practice in teacher education. The systematic review by Forsström and Munthe (2023) examined what characterized research on Nordic initial teacher education from 2010 to 2020. One finding was that the topic of "field practice" was included in 140 studies. These studies fell into three distinct subcategories: the first was studies focused on international field practice perspectives; the second was studies focused on mentoring and assessment; and the third category consisted of studies examining student teachers' teaching experiences during field practice. However, the latter subset of research primarily explored student teachers' experiences of important teaching-related concepts, such as pedagogical content knowledge (e.g., Juhler, 2016, 2018; Marie-Helena & Pernilla, 2019; Sjöberg & Nyberg, 2020) and reflective practices (e.g., Haugen et al., 2013; Toom et al., 2015; Körkkö et al., 2016; Tianen et al., 2018). None of the studies on the topic of field practice specifically examined the relationship between these experiences and self-efficacy in relation to teacher and teaching-related practices, which student teachers may doubt they have sufficient ability to enact in order to learn during field practice.

Thus, while teacher self-efficacy (i.e., self-efficacy for teaching for in-service teachers) is a well-studied field (e.g., Tschannen-Moran et al., 1998; Zee & Koomen,

2016; Morris et al., 2016), and measures of teaching self-efficacy, to some extent, have been applied to student teachers in relation to field practice. Also, related concepts, such as doubts related to developing a teacher identity, content knowledge, field practice related concerns, and reflective practices (e.g., Anspal et al., 2018, Goh & Matthews, 2011; Juhler 2016; Toom et al., 2015) have also been studied previously. However, student teachers' field practice-related self-efficacy, understood as what student teachers could have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice, is an area lacking research.

1.2. The current study

The overall aims of the study were: 1) to uncover which teacher and teaching-related practices student teachers may have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice; and 2) to discover differences and similarities between the internal (i.e., the student teachers themselves) and the external (i.e., the field-practice teachers) perspectives on this, in order to gain a first and deeper understanding of the more specific qualitative essence of student teachers' practice self-efficacy.

1.2.1. Study context

While teacher education differs across cultures and countries, it often consists of both on-campus teaching, where the academic, pedagogical, and didactical subjects are taught, and field practice, in order to acquire the profession's core skills and practices (Weisdorf, 2020). The most common teacher education appears to be a subject matter bachelor's degree, followed by an education degree at the master level, where the latter includes field practice for in-school training (Weisdorf, 2020). The Danish teacher education program differs from many other teacher education programs, as it is a four-year professional bachelor program at a university college. Similar to the other Nordic countries, it is a so-called integrated program, focusing on the coherence between field practice, didactical, and pedagogical subjects, and teaching subjects (Act on the Education and Recruitment of Teachers and Administrators of Schools of 2017; Weisdorf, 2020).

The duration of field practice and its location within the teacher education curriculum also vary across countries (Weisdorf, 2020). In Denmark, a new teacher education curriculum, which expands the field practice, was implemented from August 2023, and thus, for a number of years, two curricula will be running in the transitional period. In the "old" curriculum, field practice amounts to 30 ECTS (European Credit Transfer System) and can be made up by as many as six field placements of varying lengths (Act on the Education of Teachers of 2015). In the "new" curriculum, field practice amounts to 40 ECTS, and placements should take place in each of the four years of the program (Act on the Education of Teachers of 2023). Field practice in the first year of study, and possibly the second, should consist of a prolonged period of time, for example one-day visits combined with a full-time block of field practice (Act on the Education of Teachers of 2023,).

2. Materials and Methods

2.1. Data collection

In order to uncover the teacher and teaching-related practices student teachers may have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice, as well as any differences or similarities between internal and external perspectives, we conducted a series of interviews in the fall of 2022. For the internal perspective, we conducted two focus group interviews with first year student teachers prior to their first field practice in primary/lower secondary school. Each focus group consisted of six students. Students were selected by on-campus teachers in the subject classes of "the teacher's basic skills", which included preparing and following students through field practice. Students were selected for diversity with regard to gender, background, and age. Students were informed that participation was voluntary, and of their rights according to the European GDPR. The gender composition in one focus group was 50–50, while the other had four female students, and they varied with regards to previous education from entering teacher education directly from high school to having a bachelor's degree. Students' ages and previous experience varied, but was not recorded. For the external perspective, we conducted four single-person interviews with field-practice teachers from schools where students' field practice takes place. Field practice teachers were selected for diversity of experience, and like the students, they were informed about the voluntary nature of their participation, and their rights according to the European GDPR. Three field practice teachers were female, one was male, and they varied in their levels of experience as teachers, from one to 38 years.

All interviews were conducted using a semi-structured interview guide. Interviewees were asked for views on which teacher and teaching-related practices student teachers may have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice. If the flow of information in single interviews, or the conversation in the focus groups, haltered, the interviewer prompted with generalized statements relating to the topic in the official skills objectives for the field practice (e.g., do you think there could be something in terms of relations/connections….?). Only the overall competence categories from the skills objectives were used as prompts: didactics, classroom management, relational work (Danish Ministry of Education at: https://www.retsinformation.dk/eli/lta/2015/1068#Bil3), since there was never a need to provide more detail for interviews to resume their flow.

The focus group interviews were approximately 45 minutes long, while the single person interviews were approximately 15 minutes long. All interviews were recorded and transcribed verbatim and anonymously for analysis.

2.2. Analyses

First, we conducted a simple extraction of all statements on teacher and teaching-related practices mentioned by interviewees as potentially connected with low

(or high) self-efficacy beliefs of student teachers in field practice. These statements were organized in a matrix of informants and teacher and teaching-related practices in order to record in which interview/focus group they were mentioned.

Secondly, we conducted a simple content analysis, focused on overlap in the individual statements from the first step of extraction. In more detail, this was done by colour-coding each statement according to the overall topic of the teacher practices. For example, statements concerning having authority were given the same colour, and statements concerning contact with parents were given another colour, and so on. Subsequently, statements were reorganized by colour/overall topic, and statements that described the same teacher and teaching-related practice, with slightly varying wording, were combined to ensure uniqueness in the statements. The resulting unique practices were then plotted in what we term an agreement circle, to illustrate in a single figure the overlap (and non-overlap) of teacher and teaching-related practices across informants and informant types respectively. For this purpose, we used Fried's R code (2017), but changed it to tidyverse (v2.0.0; Wickham et al., 2019) to provide a more accessible and readable code, and adapted it to improve visual clarity. In terms of organizing the statements around the circle, we arranged them counterclockwise based on how many informants mentioned them – from most to least frequent. Informants were positioned in concentric circles, with students in the outermost circles and field-practice teachers in the innermost. In the event of statements mentioned by an equal number of informants, we determined their order based on the mentions in the outer circles, thereby making it clear whether the statement was mentioned by students. The aim was to highlight the frequency of each statement, and but also to clarify its relationship to informant types (i.e., students or field-practice teachers) and uncover potential differences, thus improving the co-occurrence circle's interpretability (the R code used is publicly available at Zenodo.org; DOI: https://zenodo.org/doi/10.5281/zenodo.10066421).

In addition, we calculated the percentages of teacher and teaching-related practices being mentioned by one, two, three (and so on) informants.

2.3 Software

R version 4.3.1 (R Core Team, 2023) was used to generate plots.

3. Results

3.1 Simple content analyses

From the simple content analysis of the interview transcriptions, we extracted a total of 63 statements related to teacher and teaching-related practices, which student teachers could have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice, according to themselves and/or their field-practice teachers. In the second step, a further content analysis revealed that some of the statements were in fact related to the same specific teacher or teaching-related practice and thus overlapping, content-wise. These content-wise

overlapping statements were combined to form unique statements; e.g., the statements[1] "develop relations with the pupils"; "to be able to form relations with children of all ages"; "speak age appropriately to all students"; "do proper relational work in a short time span"; and "remember the names of the pupils" were combined into the unique statement "develop relations with all pupils". While a statement like "make them [the pupils, red.] like me" was kept as is, as there were no similar statements. Combining the overlapping statements, reduced the number of statements on teacher and teaching-related practices to 33 unique statements (Table 1).

Almost half of the unique statements (45.5%) were mentioned by only a single informant (that is, in a single interview or focus group), while at the other end of the spectrum, only a single unique statement (3.0%) was mentioned by all six informants (Table 2).

3.2 Content analysis focused on overlap

The organization of the 33 unique statements on teacher and teaching-related practices, which student teachers could have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice, into an agreement circle, revealed that the statement that was mentioned by all 6 informants: "S7. have enough subject knowledge" (Figure 1, Table 1). A further two unique, though related, practices were mentioned by all but one informant (i.e., both student focus groups and three field-practice teachers): "S26. combination of domain knowledge and personal competence enough to be a teacher"; and "S5. carry out classroom management/ be in control of authority". Also mentioned by a majority of informants (i.e., four informants with different combinations of student focus groups and field-practice teachers) were four statements with relatively different content: "14. develop relations with all pupils"; "17. prepare good and relevant teaching where all pupils learn regardless their level"; "21. the balance between the pedagogical agenda and learning"; and "27. be able to readjust in the moment from what has been planned".

Looking across all statements mentioned by representatives of both student teachers and field-practice teachers, they have four distinct foci (Table 1, Figure A-1 in the Appendix): being prepared and being able to diverge from plans (S7, S8, S17, S27); the necessity of balancing subject knowledge and pedagogical competence for being a teacher (S6, S20, S26); taking leadership in the classroom (S5, S21, S22); and lastly, being able to teach without a field-practice teacher present (S25).

3.2.1 The internal and external perspectives

The majority of statements were put forward by either student teachers or field-practice teachers, thus providing insight into the exclusively internal and exclusively

1 Please note that interviews were done in Danish, and this accounts for some of the wordings of the statements, as we have tried to convey the form in which they were given.

Table 1: Unique statements on the teacher and teaching-related practices, which student teachers could have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice

NUMBERED STATEMENTS
S1. know what it takes to help the weaker pupils
S2. how to help pupils with special needs
S3. cope with boundary-seeking pupils
S4. be able to achieve tranquillity in class /capture the pupil's' attention
S5. carry out classroom management/ be in control of authority
S6. balance between authority/recognition and being friends
S7. have enough subject knowledge
S8. be prepared enough
S9. enough knowledge to be an authority
S10. write to/with parents
S11. be able to communicate orally with parents
S12. teach subjects that are not specialization subjects (no knowledge)
S13. make them [the pupils, red.] like me
S14. develop relations with all pupils
S15. create mutual respect (teacher/pupil)
S16. not behave in a biased way due to prior information about pupils
S17. prepare good and relevant teaching where all pupils learn regardless of their level
S18. make the pupils participate in the learning process
S19. differentiate teaching rather than just «teach myself»
S20. the balance between the pedagogical agenda and learning
S21. show enthusiasm/authenticity while teaching
S22. show initiative in shared teaching/be helpful/approach pupils and inquire
S23. get good ideas for teaching
S24. be able to manage pupil conflict
S25. be able to do it by myself (without the field practice teacher)
S26. combination of domain knowledge and personal competence enough to be a teacher
S27. be able to readjust in the moment from what has been planned
S28. let go of control and put the pupils in focus/time and place for concentration with the pupils
S29. get through everything (that has been planned)
S30. to stand in front of the pupils/be able to teach
S31. plan single lessons
S32. plan short series of lessons
S33. plan lessons for a longer course of study

Table 2: Number of statements mentioned by different numbers of informants

NO. OF UNIQUE STATEMENTS	NO. OF INFORMANTS	PROPORTION OF THE UNIQUE STATEMENTS STEMMING FROM THAT NUMBER OF INFORMANTS (%)
15	1	45.5
7	2	21.2
4	3	12.1
4	4	12.1
2	5	6.1
1	6	3.0

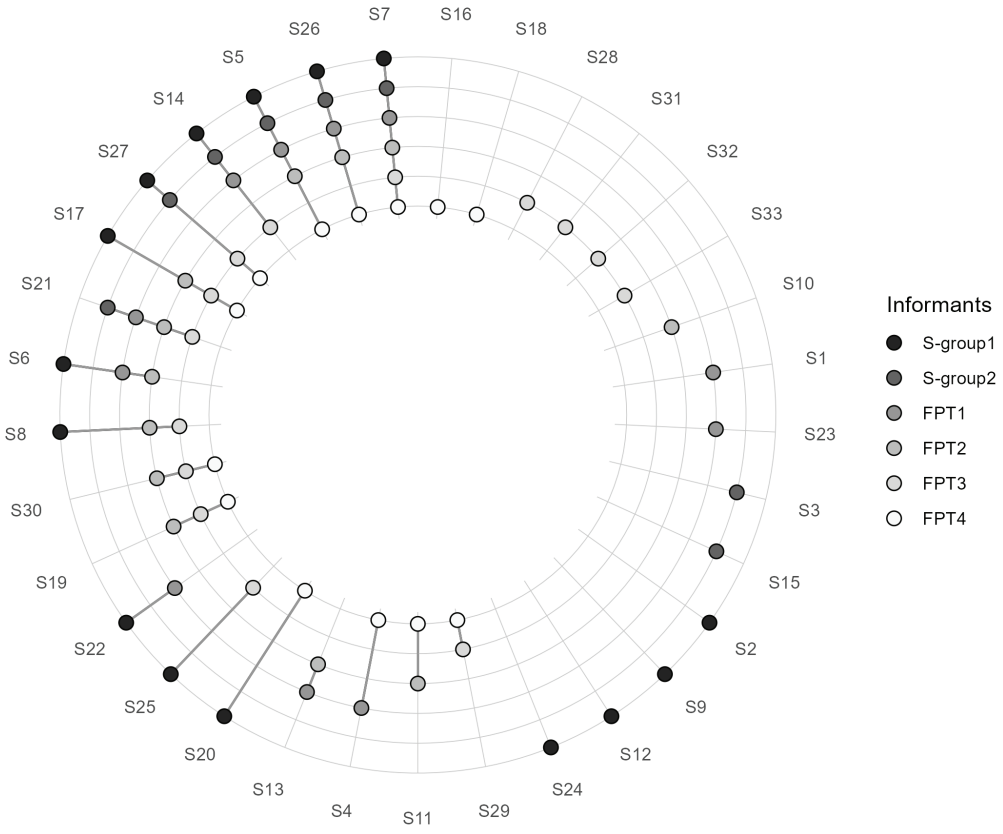


Figure 1: Teacher and teaching-related practices mentioned by informants

Note: S-groups = student focus groups (six students in each group), FPT = individual field-practice teachers

external perspectives on teacher and teaching-related practices, which student teachers could have difficulty believing they have sufficient abilities and competencies to enact/engage in while in field practice. A clear pattern of differences in the focus from the internal and external perspectives emerged (Figure 1, Table 1).

The exclusively internal student perspective was represented by statements S2, S3, S9, S12, S15, S24. Looking at the content of these statements, they were focused in two directions. Three of the statements were concerned with meeting the needs of non-average pupils and teaching what you are not specialized in: "S2. how to help pupils with special needs"; "S3. cope with boundary-seeking pupils"; and "S12. teach subjects

that are not specialization subjects (no knowledge)". The remaining three statements were all concerned with being an authority and being in charge in the classroom: "S9. enough knowledge to be an authority"; "S15. create mutual respect (teacher/pupil)"; and "S24. be able to manage pupil conflict".

The exclusively external field-practice teacher perspective was not as narrowly focused as the internal student perspective. It contained 13 statements with various foci. The main focus in the external perspective was a didactical one, represented by statements: "S23. get good ideas for the teaching"; "S29. get through everything (that has been planned)"; "S31. plan single lessons"; "S32. plan short series of lessons"; and "S33. plan lessons for a longer course of study". Another notable focus was facilitation of learning for the benefit of the pupils with statements: "S4. be able to achieve tranquillity in class /catch the pupils' attention"; "S18. make the pupils participate in the learning process"; and "S28. let go of control and put the pupils in focus/time and place for concentration with the pupils". Less pronounced was a focus on pupils' perception of the student teacher and vice versa: "S13. make them [the pupils, red.] like me [the student teacher, red.]"; and "S16. not behave in a biased way due to prior information about pupils". There was also a focus on communication with parents: "S10. write to/with parents" and "S11. be able to communicate orally with parents". A single statement was similar to the first focus area of the internal student perspective, namely facilitating learning for the weaker pupils: "S1. know what it takes to help the weaker pupils". None were similar to the second focus of the internal student perspective of being respected and naturally authoritative.

4. Discussion and Future Directions

With the current study, we have uncovered a range of teacher and teaching-related practices in which student teachers may be doubtful about having the necessary abilities and skills to engage in while in field practice. We have also uncovered both similarities and differences in the teacher and teaching-related activities brought forward by the student teachers themselves from the internal perspective and the field-practice teachers from the external perspective.

Both student teachers and field-practice teachers mentioned four distinct areas of teacher and teaching-related activities where student teachers doubt they have what it takes to engage in the field practice, i.e., areas where the internal and external perspectives are in congruence. The first area was didactical, both in the sense of prior preparation and the ability to diverge from plans in the teaching situation (i.e., be adaptable), as expressed in S7, S8, S17, S27. That adaptability is mentioned as an area of doubt by both student teachers and field practice teachers is not surprising, as this is an advanced competence developed through practice and on the basis of didactical, pedagogical, and subject knowledge (Cho, 2022). The second area was also didactical, but more concerned with the "totality" of the didactical teacher competence, as a balancing of subject knowledge and pedagogical competence, as expressed in S6, S20, S26. This was in line with Goh and Matthews (2011), who also found this to be an area of concern to student teachers, though in terms of concerns related to field practice

rather than doubt in own abilities, as in the present study. The third area was concerned with taking leadership in the classroom, as expressed in S5, S21, S22, and thus an area more concerned with the pedagogical side of teaching, as well as teacher personality or teacher style. This finding was also in line with findings of Goh and Matthews (2011) on student teacher concerns related to field practice, as well as Anspal et al.'s (2018) finding that the relationship between self-conception and the professional role they were entering as student teachers is a source of tension. The fourth area, mentioned both by field-practice teachers and student teachers, was being able to teach without a field-practice teacher present (S25). This thus can be seen as an umbrella, which covers all the previously mentioned areas, as this would entail all of these, and could also be seen as similar to the already mentioned finding of Anspal and colleagues (2018) on the tensions created by the relationship between self-conception and the new professional role.

Doubting oneself, being concerned with standing out among fellow student teachers, or "falling short" in the eyes of field-practice teachers and pupils, emerges as a theme from the student statements, specifically: "S9. enough knowledge to be an authority"; "S12. teach subjects that are not specialization subjects (no knowledge)"; and "S25. be able to do it by myself (without the field-practice teacher)". We wonder, if these statements could reflect a premonition of the practice shock, which has also been documented among student teachers (Meijer, de Graaf & Meirink, 2011). Or could this stem from a fear of being revealed as less than adequate; i.e., could it be an expression of imposter syndrome (Clance, & Imes, 1978)? Imposter syndrome has been documented among university students, internationally (Pákozdy et al., 2023) as well as in the Danish context (Laursen & Sandberg, 2022), and also specifically among preservice teachers (LaPalme et al., 2022). LaPalme et al. (2022) showed that a pathway towards mitigating imposter syndrome for preservice teachers was emotion regulation. This aligns well with Bandura's (1997) suggestion that affective states are a source of self-efficacy, and that affective processes are mediated by self-efficacy and could thus point to an additional avenue of future research building on the current study, i.e., the dual relationship between the experience of imposter syndrome among student teachers and their self-efficacy for engaging in teacher and teaching-related practices during field practice in schools.

The emerging internal themes (student teachers) of meeting the needs of non-average pupils, being an authority and being in charge in the classroom differ substantially from the emerging external themes (field-practice teachers) of didactical issues, namely the mutual perception of pupils and student teachers and engaging with parents. The field-practice teachers specifically point out the ability to engage and enact as teachers towards parents, as seen in "S10. write to/with parents" and "S11. be able to communicate orally with parents" as an area where they think the student teachers will doubt their abilities. This parental focus, however, does not show up among student teacher statements. Why is that? It is beyond the scope of the current study to provide the answer, but an influencing factor for these differences, between the

internal perspective of the student teachers and the external perspective of field practice teachers, is the teacher education curriculum on field practice. Student teachers do not take part in the full range of teaching practices during their field practice. In the new curriculum at the university college in the study, student teachers get to "wet their toes before they plunge into the pool of field practice". Thus, they will, in the first year, be at the practice school for a prolonged period of time with regular one-day school visits, followed by a three-week full-time field practice, and yet another prolonged period of time with regular one-day school visits. This means that student teachers are not an integral part of the teams of teachers teaching the same grade, where the teachers collaborate on planning, use of materials and teaching methods, and reflect together. The time-wise restrictions on the field-practice also mean that student teachers are legitimate peripheral participants in the ongoing home-school partnership (Lave & Wenger, 1991), and they usually communicate with parents via their field practice teachers, rather than practicing communication themselves. The result may well be that the field practice is focused on teaching and the performance of the student teacher, rather than the ongoing collaborative nature of teaching practice, in which relations with pupils, colleagues, and parents are central parts. This position is supported by Goh et al.'s (2017) study, which suggested that the learning environment of contemporary novice teachers invites an understanding of teacher competency in terms of right and wrong, thus leaving less room for contradictions, tensions, and complexities as part and parcel of teaching practice. Thus, the curriculum and context itself contribute to student teachers believing a somewhat naïve theory (Savion, 2009) of what constitutes teaching practice, rather than developing a more thorough understanding of what it entails to be a teacher to form the basis of their own teaching practice. We are, of course, aware that the perspectives of the student teachers and the field-practice teachers differ by nature. One is a personal internal understanding. The other is an external understanding, and though built on having previously been a student, it also incorporates the professional understanding and experience of the full collaborative teacher role in the situated complexity of the classroom and school context (Mardahl-Hansen, 2018). Thus, beliefs about what it takes to engage in/enact the profession's core practices must differ, as they emerge from different levels of experience, even if the teaching practices have immanent characteristics (Bandura, 1977; 1997), or the field-practice context affords certain practices (Gibson, 1979).

While this study contributes new insights into both themes and specific practices that may create uncertainty among student teachers, it also has limitations. The findings were based on focus group interviews and single person interviews, with a total of twelve students and four field practice teachers. Also, while interviews were ended when no new information could be provided by interviewees, we might have missed some practices that would be relevant to consider in terms of student teacher field practice self-efficacy. On the other hand, our use of both an internal perspective (i.e., interviewing students) and an external perspective (i.e., interviewing field practice teachers) enhances the study's scope in identifying a wide repertoire of challenging

teaching practices. Either way, although the study has made an important contribution in identifying specific teacher and teaching-related practices that student teachers might feel uncertain about engaging in, in order to learn and eventually master them, the identified practices stem from only two groups of field-practice stakeholders.

Given the limitations, there are multiple ways future research could extend our findings. A natural first step would be to include a larger number of interviewees and a more diverse pool of field-practice stakeholders to discover further unidentified teaching practices, which could be relevant for student teacher field practice self-efficacy, for example, university college teachers involved in field practice. Previous research has not specifically discovered which specific teacher and teaching-related practices student teachers and their field-practice teachers would point out as being issues regarding self-efficacy. Instead, previous research has focused on more general teacher self-efficacy, or teaching subject-specific self-efficacy (e.g., Enochs & Riggs, 1990; Flores, 2015; Li & Zhang, 2000). Thus, another promising direction for future research could be to build on the current study, and an expansion of it, to a quantitative assessment of student teachers' levels of confidence in enacting these practices during field practice. In this sense, what has yet to be investigated is the actual degree of confidence in having sufficient abilities, skills, and competence to engage in these practices in field practice at different points in the teacher education, or what might be termed self-efficacy for field practice, or practice self-efficacy in professional education. Such a measure could help to assess and enhance student teachers' training during their education.

Previous research has shown that over the course of a year in a Master of Education program, where most of the year was spent in school placements, the student teachers' teacher self-efficacy did increase significantly (though not substantially) from the start to the end of the year (Woolfolk Hoy & Spero, 2005). However, it was also found that from the end of the placement year to the end of their first year of teaching (after graduating), the now teachers' general teacher self-efficacy decreased back to the level it was before the year of school placement. Thus, teacher self-efficacy from within the time of education to after having been working in the teaching profession for a year fluctuated, but showed no stable increase. Other research has indicated that student teachers' meeting with teaching through field practice can cause a decline in their self-efficacy (Anspal et al., 2018), as it was found that tensions arising from the relationship between self-conception and the professional role increased for some student teachers as they progressed in their education. Possibly an increase in practice self-efficacy during teacher education is helpful for students while they are still in the program, since they might more easily engage in the various practices in field-practice. However, once they meet the "full" job context with its many demands as graduated teachers, self-efficacy might drop back down. To elucidate these questions, longitudinal studies focusing on student teachers' self-efficaciousness of engaging in the specific teacher and teaching-related practices while in field practice is needed to investigate whether practice self-efficacy changes as a result of the field practice experience,

and whether such changes depend on the time-wise placement of the field practice in the teacher education program (i.e., early or late). Thus, longitudinal studies assessing practice self-efficacy prior to field practice, during field practice, after field practice, throughout the teacher education and beyond, will provide entirely new knowledge on the reciprocal relationship between practical experience and self-efficacy (Bandura, 1997), in the particular context of field practice in teacher education. Whether practice self-efficaciousness is retained if achieved during education, or in the transition to work-life of the new teachers, is a promising course of future research on practice self-efficacy in professional education. Such future studies would not only be in line with the gap in Nordic research on field practice in initial teacher education (Forsström & Munthe, 2023), but also contribute to the gap in self-efficacy research, as no previous studies have specifically targeted field practice self-efficacy in teacher education, understood as the doubts students may have in relation to engaging in teacher and teaching-related practices, in order to learn and eventually master them.

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Disclosure statement

The authors report there are no competing interests to declare.

Ethics

No ethical approval is needed in Denmark for research involving only survey data. Participating students were informed of their right to withdraw from the study at any time prior to data anonymization, as well as all other rights, and how their data would be treated according to current European data protection regulations (see also section 2.1). In addition, we only analysed interview transcripts that were anonymized, and we have not quoted individual persons in the article.

Data availability

The data matrix used to plot the agreement circle in Figure 1 is available at Zenodo.org, DOI: https://zenodo.org/doi/10.5281/zenodo.10066446.

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REFERENCES

Act on the Education and Recruitment of Teachers and Administrators of 2017. (2017). *Act on the education and recruitment of teachers and administrators of preschools, compulsory schools and upper secondary schools, No 87/2008*. https://www.government.is/publications/legislation/lex/2017/11/22/Act-on-the-educationand-recruitment-of-teachers-and-administrators-of-preschools-compulsory-schools-and-uppersecondary-schools-No-87-2008/

Act on the Education of Teachers of 2015. (2015). *Bekendtgørelse om uddannelsen til professionsbachelor som lærer i folkeskolen, BEK nr. 1068 af 08/09/2015*. https://www.retsinformation.dk/eli/lta/2015/1068

Act on the Education of Teachers of 2023. (2023). *Bekendtgørelse om uddannelsen til professionsbachelor som lærer i folkeskolen, BEK nr. 374 af 29/3/2023*. https://www.retsinformation.dk/eli/lta/2023/374

Alsup, J. (2005). *Teacher identity discourses: Negotiating personal and professional spaces*. Routledge. https://doi.org/10.4324/9781410617286

Anspal, T., Leijen, Ä., & Löfström, E. (2018). Tensions and the teacher's role in student teacher identity development in primary and subject teacher curricula. *Scandinavian Journal of Educational Research, 63*(5), 679–695. https://doi.org/10.1080/00313831.2017.1420688

Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavior change. *Psychological Review, 84*(2), 191–215. https://doi.org/10.1037/0033-295X.84.2.191

Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman and Company.

Brown, A. L., Myers, J., & Collins, D. (2021). How pre-service teachers' sense of teaching efficacy and preparedness to teach impact performance during student teaching. *Educational Studies, 47*(1), 38–58. https://doi.org/10.1080/03055698.2019.1651696

Caires, S., & Almeida, L. S. (2005). Teaching practice in initial teacher education: Its impact on student teachers' professional skills and development. *Journal of Education for Teaching, 31*(2), 111–120. https://doi.org/10.1080/02607470500127236

Cho, K. (2022). *Developing teaching adaptability in pre-service teachers using practice-based teacher education*. [Doctoral dissertation, West Virginia University]. The Research Repository @ WVU. https://researchrepository.wvu.edu/etd/11596

Clance, P. R., & Imes, S. (1978). The imposter phenomenon in high achieving women: Dynamics and therapeutic interventions. *Psychotherapy: Theory, Research & Practice, 15*(3), 1–8. https://doi.org/10.1037/h0086006

Enochs, L. G., & Riggs, I. M. (1990). Further development of an elementary science teaching efficacy belief instrument: A preservice elementary scale. *School Science and Mathematics, 90*(8), 694–706. https://doi.org/10.1111/j.1949-8594.1990.tb12048.x

Forsström, S. E., & Munthe, E. (2023). What characterizes Nordic research on initial teacher education: A systematic scoping review. *Nordic Studies in Education, 43*(3), 241–259. https://doi.org/10.23865/nse.v43.5588

Fried, E. I. (2017). The 52 symptoms of major depression: Lack of content overlap among seven common depression scales. *Journal of Affective Disorders, 208*, 191–197. https://doi.org/10.1016/j.jad.2016.10.019

Flores, I. M. (2015). Developing preservice teachers' self-efficacy through field-based science teaching practice with elementary students. *Research in Higher Education Journal, 27*. https://files.eric.ed.gov/fulltext/EJ1056173.pdf

Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.

Gibson, S., & Dembo, M. (1984). Teacher efficacy: A construct validation. *Journal of Educational Psychology, 76*, 569–582. https://doi.org/10.1037/0022-0663.76.4.569

Goh, P., & Matthews, B. (2011). Listening to the concerns of student teachers in Malaysia during teaching practice. *Australian Journal of Teacher Education, 36*, 92–103. http://dx.doi.org/10.14221/ajte.2011v36n3.2

Goh, P. S. C., Yusuf, Q., & Kung, T. W. (2017) Lived experience: Perceptions of competency of novice teachers. *International Journal of Instruction, 10*(1), 21–36. http://dx.doi.org/10.12973/iji.2017.1012a

Haugan, J. A., Moen, T., & Karlsdóttir, R. (2013). Exploration of Norwegian student teachers' reflective mediation during internships. *Reflective Practice, 14*(2), 226–239. https://doi.org/10.1080/14623943.2012.749233

Hoy, W. K., & Woolfolk, A. E. (1990). Socialization of student teachers. *American Educational Research Journal, 27*, 279–300. https://doi.org/10.3102/00028312027002

Juhler, M. V. (2016). The use of lesson study combined with content representation in the planning of physics lessons during field practice to develop pedagogical content knowledge. *Journal of Science Teacher Education, 27*(5), 533–553. https://doi.org/10.1007/s10972-016-9473-4

Juhler, M. V. (2018). Pre-service teachers' reflections on teaching a physics lesson: How does lesson study and content representation affect pre-service teachers' potential to start developing PCK during reflections on a physics lesson. *Nordic Studies in Science Education, 14*(1), 22–36. https://doi.org/10.5617/nordina.2433

Körkkö, M., Kyrö-Ämmälä, O., & Turunen, T. (2016). Professional development through reflection in teacher education. *Teaching and Teacher Education, 55*, 198–206. https://doi.org/10.1016/j.tate.2016.01.014

Laursen, M. G., & Sandberg, K. (2022). Impostor-fænomen blandt fremspirende voksne: En kvalitativ interviewundersøgelse af oplevelsen og håndteringen af impostor-fænomen. *Qualitative Studies, 7*(1), 175–200. https://doi.org/10.7146/qs.v7i1.133072

Lave, J., & Wenger, E. (1991). *Situated learning: Legitimate peripheral participation.* Cambridge University Press.

LaPalme, M., Luo, P., Cipriano, C., & Brackett, M. (2022). Imposter syndrome among pre-service educators and the importance of emotion regulation. *Frontiers in Psychology, 13*. https://doi.org/10.3389/fpsyg.2022.838575

Li, X., & Zhang, M. (2000). *Effects of early field experiences on preservice teachers' efficacy beliefs: A pilot study.* [Conference presentation] Annual Meeting of the American Educational Research Association. https://files.eric.ed.gov/fulltext/ED444973.pdf

Mardahl-Hansen, T. (2019). Teaching as a social practice. *Nordic Psychology, 71*(1), 3–16. https://doi.org/10.1080/19012276.2018.1457451

Marie-Helene, Z. N., & Pernilla, N. (2019). From pedagogical knowledge to pedagogical content knowledge: Development in mentor and student-teacher group conversations. *International Journal of Learning, Teaching and Educational Research, 18*(10), 233–247. https://doi.org/10.26803/ijlter.18.10.15

Meijer, P.C., de Graaf, G., & Meirink, J. (2011). Key experiences in student teachers' development. *Teachers and Teaching, 17*(1), 115–129. https://doi.org/10.1080/13540602.2011.538502

Mok, S. Y., Rupp, D., & Holzberger, D. (2023). What kind of individual support activities in interventions foster pre-service and beginning teachers' self-efficacy? A meta-analysis. *Educational Research Review, 40*, 100552. https://doi.org/10.1016/j.edurev.2023.100552

Morris, D.B., Usher, E.L., & Chen, J.A. (2017). Reconceptualizing the sources of teaching self-efficacy: A critical review of emerging literature. *Educational Psychology Review, 29*, 795–833. https://doi.org/10.1007/s10648-016-9378-y

Pákozdy, C., Askew, J., Dyer, J., Gately, P., Martin, L., Mavor, K.I., & Brown, G.R. (2023). The imposter phenomenon and its relationship with self-efficacy, perfectionism and happiness in university students. *Current Psychology*. https://doi.org/10.1007/s12144-023-04672-4

R Core Team (2023). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Savion, L. (2009). Clinging to discredited beliefs: The larger cognitive story. *Journal of the Scholarship of Teaching and Learning, 9*(1), 81–92.

Sjöberg, M., & Nyberg, E. (2020). Professional knowledge for teaching in student teachers' conversations about field experiences. *Journal of Science Teacher Education, 31*(2), 226–244. https://doi.org/10.1080/1046560X.2019.1688533

Tiainen, O., Korkeamäki, R. L., & Dreher, M. J. (2018). Becoming reflective practitioners: A case study of three beginning pre-service teachers. *Scandinavian Journal of Educational Research, 62*(4), 586–600. https://doi.org/10.1080/00313831.2016.1258673

Toom, A., Husu, J., & Patrikainen, S. (2015). Student teachers' patterns of reflection in the context of teaching practice. *European Journal of Teacher Education, 38*(3), 320–340. https://doi.org/10.1080/02619768.2014.943731

Tschannen-Moran, M., Woolfolk Hoy, A., & Hoy, W. K. (1998). Teacher efficacy: Its meaning and measure. *Review of Educational Research, 68*(2), 202–248. https://doi.org/10.2307/1170754

Weisdorf, A. K. (2020). *Læreruddannelsen i globalt perspektiv: Et komparativt studie af læreruddannelsen i Danmark, England, Finland, Holland, New Zealand, Norge, Ontario, Singapore, Sverige og Tyskland.* Danske Professionshøjskoler. https://danskeprofessionshøjskoler.dk/wp-content/uploads/2022/01/Laereruddannelsen-i-globalt-perspektiv.-Et-komparativt-studie.pdf

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., ... Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Woolfolk Hoy, A., & Burke Spero, R. (2005). Changes in teacher efficacy during the early years of teaching: A comparison of four measures. *Teacher and Teacher Education, 21*, 343–356. https://psycnet.apa.org/doi/10.1016/j.tate.2005.01.007

Zee, M., & Koomen, H. M. Y. (2016). Teacher self-efficacy and its effects on classroom processes, student academic adjustment, and teacher well-being: A synthesis of 40 years of research. *Review of Educational Research, 86*(4), 981–1015. https://doi.org/10.3102/0034654308325693

Appendix. Additional results

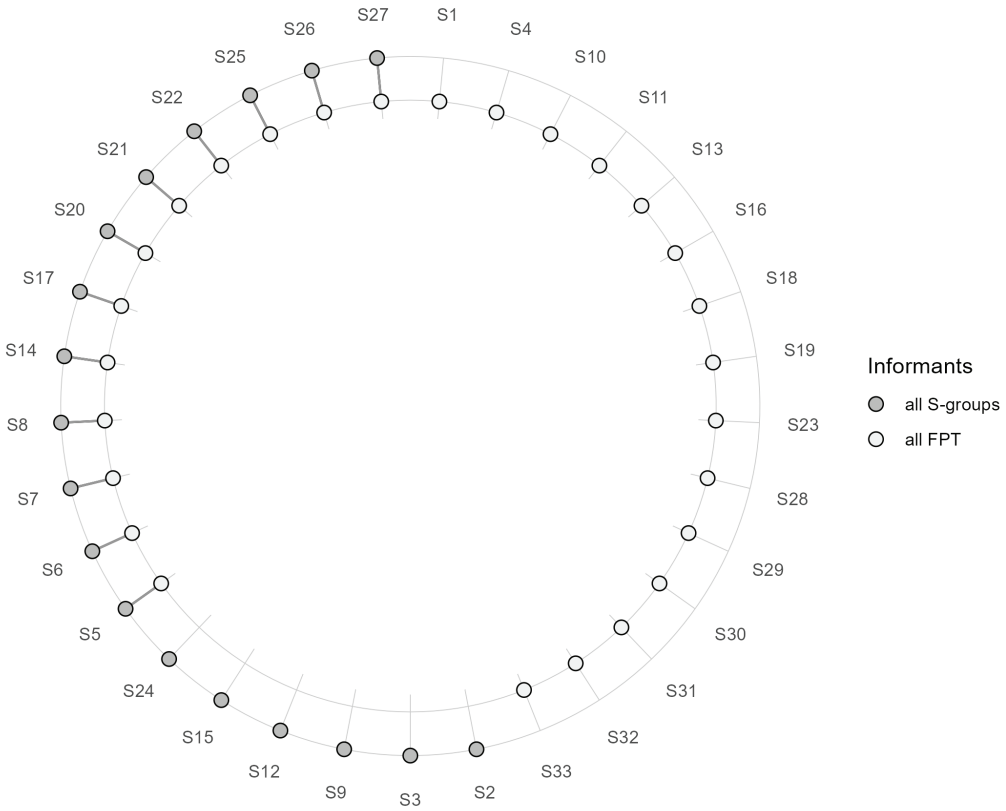


Figure A-1: Teacher and teaching-related practices mentioned by category of informants

Note: S-groups = across student focus groups, FPT = across field practice teachers